

HAWASSA UNIVERSITY MODEL EXAM

Model exam 2 for Bed and BSc chemistry students

- Student A was doing an experiment to determine the surface tension of an organic liquid using drop count method. The student used an instrument called stalagmometer and counted the number of drops from the fixed (equal) volume of water and organic liquids. Three measurements (trials) were done for each liquid and the results are summarized in table below.

Trial	Number of drops counted from the fixed volume of liquid	
	water	Organic liquid
1	36	55
2	34	54
3	36	56

What is the surface tension of organic liquid if the densities of water and the organic liquid are 0.996 and 0.80 g cm^{-3} and the surface tension of water is $7.2 \times 10^{-2} \text{ N m}^{-1}$?

- A) $7.2 \times 10^{-2} \text{ N m}^{-1}$
 B) $3.68 \times 10^{-2} \text{ Nm}^{-1}$
 C) $2.4 \times 10^{-3} \text{ Nm}^{-1}$
 D) $6.3 \times 10^{-3} \text{ Nm}^{-1}$

Questions from 2-4 is based on the data shown in the table. Student B optimized the CO molecule at B3LYP/STO-3G computational level of theory to analyze frontal molecular orbitals of the molecule. The Eigen values, molecular orbital coefficients and symmetry designations for molecular orbitals for part of CO molecule are printed out from Gaussian 09 as shown in table below.

			6	7	8	9	10
			0	0	V	V	V
Eigenvalues --			-0.33766	-0.21891	0.01182	0.01182	0.53011
1	1	C 1S	0.00000	-0.18041	0.00000	0.00000	-0.11774
2		2S	0.00000	0.77016	0.00000	0.00000	0.80712
3		2PX	0.00000	0.00000	0.90604	0.00000	0.00000
4		2PY	0.47866	0.00000	0.00000	0.90604	0.00000
5		2PZ	0.00000	-0.53127	0.00000	0.00000	1.16453
6	2	0 1S	0.00000	-0.00657	0.00000	0.00000	0.11971
7		2S	0.00000	0.04737	0.00000	0.00000	-0.91156
8		2PX	0.00000	0.00000	-0.66486	0.00000	0.00000
9		2PY	0.77973	0.00000	0.00000	-0.66486	0.00000
10		2PZ	0.00000	0.47423	0.00000	0.00000	0.92443

- Which molecular orbital is HOMO?

- A) Molecular orbital 7
 B) Molecular orbital 8
 C) Molecular orbital 9
 D) Molecular orbital 10

3. Which of the following is true about the molecular orbital 9?
 - A) It is π type
 - B) It is π^* type
 - C) It is n type
 - D) It is σ type
4. Electronic transition from molecular orbital 6 to 8 is _____ type
 - A) $n \rightarrow \pi$
 - B) $n \rightarrow \pi^*$
 - C) $\pi \rightarrow \pi^*$
 - D) $\pi \rightarrow \pi$
5. E° value for the Mn^{+3}/Mn^{+2} couple is positive (+1.5 V) whereas that of Cr^{+3}/Cr^{+2} is negative (-0.4 V). The reason for this variation is
 - A) Mn^{+2} is more stable than Mn^{+3} due to half-filled d-orbitals ($3d^5$), whereas Cr^{+3} is more stable than Cr^{+2} due to half-filled orbitals
 - B. the maximum oxidation state exhibited by manganese
 - C. The greater size of chromium compared to manganese
 - D. Mn^{+2} is less stable than Mn^{+3} whereas Cr^{+3} is less stable than Cr^{+2} in aqueous solutions.
6. Which one of the following characteristics of the transition metals is associated with higher catalytic activity?
 - A. High enthalpy of atomization
 - B. Paramagnetic behavior
 - C. Colour of hydrate ions
 - D. Variable oxidation states
7. Which of the following is wrong about either orthotitanic acid, $TiO_2 \cdot 2H_2O$ or metatitanic acid, $TiO \cdot H_2O$?
 - A. $TiO_2 \cdot 2H_2O$ can be prepared by the reaction of titanyl sulphate ($TiOSO_4$) with NaOH.
 - B. Although $TiO_2 \cdot 2H_2O$ displays amphoteric behavior; it readily dissolves in H_2SO_4 but is insoluble in NaOH, indicating that it has a more basic character.
 - C. Metatitanic acid, $TiO_2 \cdot H_2O$ may be prepared by boiling $TiO_2 \cdot 2H_2O$ in NaOH
 - D. Although it is soluble in H_2SO_4 , metatitanic acid, $TiO \cdot H_2O$ is insoluble in NaOH.
8. All of the following are correct except
 - A. Titanium(IV) compounds can be reduced to titanium (III) using Zn as a reducing agent.
 - B. Ti(IV) compounds can be identified using the reaction: $TiO_2^+ + H_2O_2 \rightarrow Ti(O_2) + H_2O$.
 - C. Metatitanic acid, $TiO_2 \cdot H_2O$ is produced when a $TiOSO_4$ solution reacts with $(NH_4)_2S$.
 - D. $TiO_2 \cdot 2H_2O$ and $TiO_2 \cdot H_2O$ may be distinguished by their solubility with H_2SO_4
9. Which of the following is wrong about chromium compounds?
 - A. The oxidation of Cr(III) to Cr(VI) can be brought about both in basic & acidic solution
 - B. Cr(III) can be oxidized to Cr(V) by H_2O_2 in alkaline solution as

$$2[Cr(OH)_6]^{3-} + 3H_2O_2 \rightarrow 2CrO_4^{2-} + 8H_2O + 2OH^-$$
 - C. When $H_2S(aq)$ is added to K_2CrO_7 , the colour changes from orange to green, indicating the oxidizing property of K_2CrO_7 .
 - D. When $H_2S(aq)$ is added to K_2CrO_7 , the colour changes from orange to green, indicating the reducing property of K_2CrO_7 .
10. $CuSO_4 \cdot 5H_2O$ is blue is colored because
 - A. It contains water of crystallization.
 - B. SO_4^{2-} ions absorb red light.

- C. Cu^{2+} ions absorb orange red light.
- D. Cu^{2+} ions absorb all colours except red from the white light
11. Among the following reactions, N_2 is not generated by heating:
- A. $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$
- B. $\text{NH}_4\text{Cl} + \text{NaNO}_2$
- C. $\text{NH}_4\text{Cl} + \text{CaO}$
- D. $\text{Ba}(\text{N}_3)_2$
12. KMnO_4 acts as oxidizing agent in alkaline medium. When alkaline KMnO_4 is treated with KI , iodide ion is oxidized to _____
- A. I_2 B. IO^- C. IO_2^- D. IO_3^-
13. Which of the following statements is not correct?
- A. Copper liberates hydrogen from acids
- B. In its higher oxidation states, manganese forms stable compound with oxygen and fluorine
- C. Mn^{3+} and Co^{3+} are oxidizing agents in aqueous solution
- D. Ti^{2+} and Cr^{2+} are reducing agents in aqueous solution
14. In the laboratory, manganese (II) salt is oxidized to permanganate ion in aqueous solution by
- A. Hydrogen peroxide
- B. Conc. Nitric acid
- C. peroxydisulfate
- D. dichromate
15. In the group of alkaline earth metals, which elements do not give flame color to the Bunsen flame ?
- A. Calcium and strontium B. Barium and strontium C. Beryllium and magnesium D. Calcium and magnesium
16. Which of the following is not correct about alkaline earth metals?
- a. All the alkaline earth metals except beryllium react with water.
- b. Alkaline earth metals react with acids to produce dihydrogen in the form of H_2 gas.
- c. Radium and barium are the least reactive whereas beryllium is the best.
- d. Alkaline earth metals dissolve in liquid ammonia to give solutions that contain two solvated electrons per metal atom.
17. Which of the following is not correct about the test of Ca^{2+} in a sample?
- I. The content of Ca^{2+} ions can be proved by the addition of ammonium oxalate, $(\text{NH}_4)_2\text{C}_2\text{O}_4$.
- II. If Ca^{2+} is present, a white precipitate of calcium oxalate (CaC_2O_4) occurs in $(\text{NH}_4)_2\text{C}_2\text{O}_4$
- III. The amount of CaC_2O_4 precipitate is proportional to the amount of Ca^{2+} ions in the solution.
- A. I and II B. II and III C. I and III D. I, II and III
18. When a solution of potassium iodide is added to a solution of lead nitrate taken in a test tube, the precipitation of a yellowish solid is observed. The net ionic equation is
- A. $\text{KI} + \text{Pb}(\text{NO}_3)_2 \rightarrow \text{PbI}_2 + \text{KNO}_3$ C. $\text{PbI}_2 + \text{KNO}_3 \rightarrow \text{Pb}(\text{NO}_3)_2 + \text{KI}$
- B. $\text{Pb}^{2+} + 2\text{I}^- \rightarrow \text{PbI}_2$ D. $\text{Pb}^{2+} + 2\text{NO}_3^- \rightarrow \text{Pb}(\text{NO}_3)_2$
19. Na imparts yellow colour to Bunsen flame because of
- A. Low ionisation potential
- B. Sensitivity

- C. Sublimation
 D. **Extra energy being absorbed and emitted as visible radiation.**

20. Consider the oxides of Group IIIA: I. B_2O_3 II. Al_2O_3 III. Ga_2O_3 IV. In_2O_3

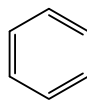
Which shows both amphoteric and acidic oxides?

- A. **I and II** B. II and III C. III and IV D. II and IV

21. Identify the correct statement about **benzene** and **cyclooctatetraene**.

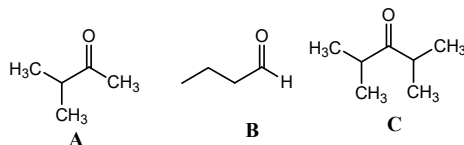


Cyclooctatetraene

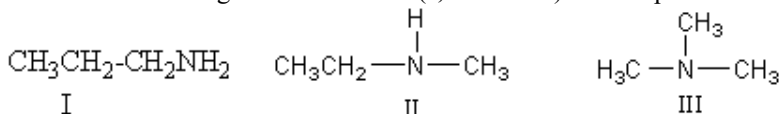


Benzene

- A. Benzene has a bent shape and cyclooctatetraene is planar.
 B. Benzene has non-delocalizing six pi-electrons and cyclooctatetraene has eight delocalizing pi-electrons.
 C. Benzene readily undergoes addition reaction whereas cyclooctatetraene undergoes electrophilic substitution reaction.
 D. Cyclooctatetraene is non-aromatic compound, and readily decolorize $KMnO_4$ and Br_2 at room temperature whereas benzene doesn't decolorize bromine at room temperature.
 E. Both compounds belong to the same family; namely aromatic compounds
22. Nucleophilic addition is characteristic reaction of ketones and aldehydes. Given compounds (A to C), their reactivity orders toward nucleophilic addition reaction are

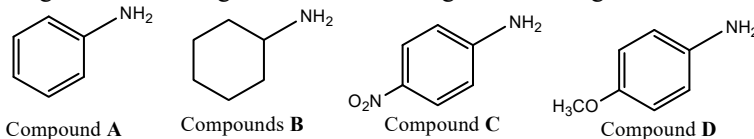


- A. $A > B > C$ B. $B > C > A$ C. $C > B > A$ D. $B > A > C$ E. All have the same rate of reactivity
23. Addition of **Grignard reagent** (CH_3CH_2-MgBr) to formaldehyde, followed by protonation gives
- A. Secondary alcohol B. Primary alcohol C. Aldehyde D. Ketones E. None of the above
24. When the **boiling points** of the following isomeric amines (**I, II and III**) are compared



- A. $I > II > III$ B. $III > II > I$ C. $II > III > I$ D. $I < II < III$ E. None of the above

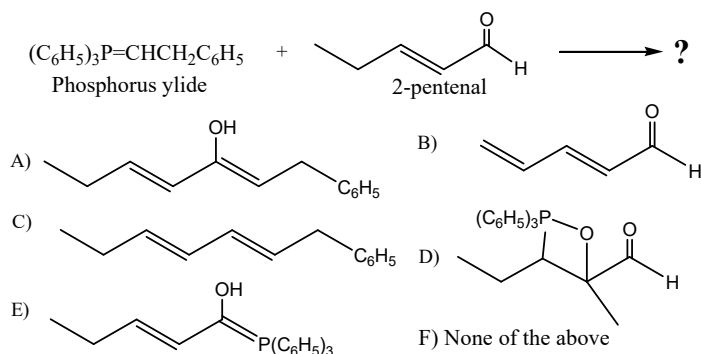
25. When the following bases are arranged in their increasing or decreasing order of basicity



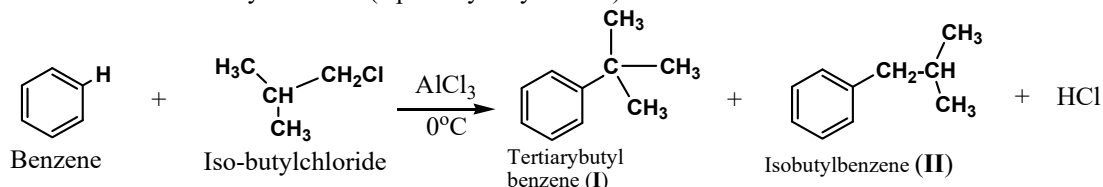
- A. $A > B > C > D$ B. $A < B < C < D$ C. $B > C > A > D$ D. $B > D > A > C$ E. All have equal basicity

26. Identify the **wrong** statement about **aldehydes** and **ketones**.

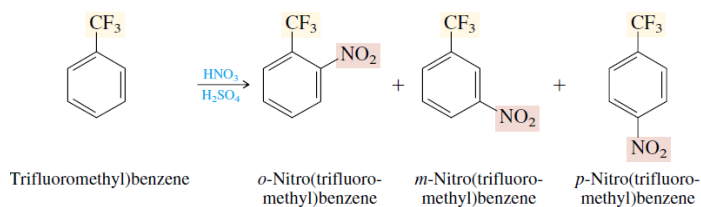
- A. Aldehydes are more reactive than ketones towards nucleophilic addition reaction.
 B. Their solubility in water decreases with molecular weight but boiling point increases with molecular weight
 C. Aldehydes and ketones can occur naturally and also play many important biological roles in living cells
 D. Ozonolysis of 2-butene gives two moles of acetaldehyde
 E. None of the above
27. Which one of the following statements is **true** about preparation and oxidation-reduction of ketones and aldehydes?
- A. Reduction of aldehydes and ketones in the presence of reducing reagents such as $NaBH_4$ gives secondary and primary alcohols, respectively
 B. Reaction of ketones with Grignard reagent ($RMgBr$) in ether gives secondary alcohols
 C. Oxidation of secondary alcohols can be used to prepare aldehydes
 D. Aldehydes can be oxidized with sodium (or potassium) dichromate in acidic medium to form carboxylic acids containing the same number of carbon atoms
 E. Redox reactions do not exist in biological systems (or living cells)
28. **Wittig reaction** is one of the common reactions of aldehydes and ketones. Which one of the following compound is the major product of the reaction shown below?



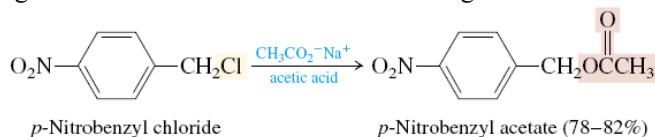
29. **Friedel-Crafts alkylation** of benzene in the presence of alkyl halides and AlCl_3 (Lewis acid catalyst) yields **alkyl benzene** via electrophilic aromatic substitution reaction. Which one of the following statements is true about reaction of benzene with isobutyl chloride (a primary alkyl halide)?



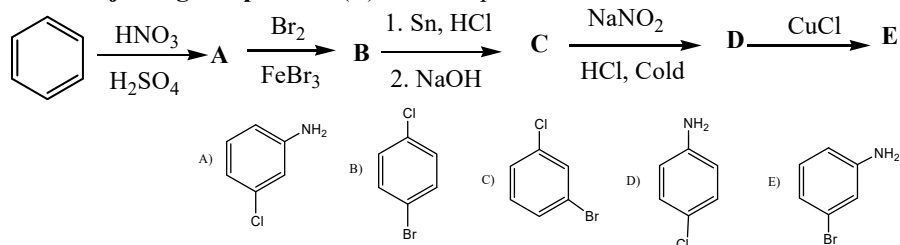
- A. The reaction yields tertiary butyl benzene (I) as a major product and isobutyl benzene (II) as minor product
 B. The reaction yields isobutyl benzene (II) as major product and tertiary butyl benzene (I) as minor product
 C. The reaction yields both isobutyl benzene (II) and tertiary butyl benzene (I) in equal amount
 D. It is difficult to decide E. None of the above
30. Nitration of trifluoro methyl benzene is a typical example of electrophilic aromatic substitution. Given trifluoro methyl benzene as starting material, the **major product** of a nitration reaction in the presence of sulfuric acid (H_2SO_4) is



- A. *O*-nitro(trifluoromethyl)benzene
 B. *p*-nitro(trifluoromethyl)benzene
 C. *m*-nitro(trifluoromethyl)benzene
 D. *O*-nitro(trifluoromethyl)benzene and *p*-nitro(trifluoromethyl)benzene
 E. None of the above
31. Which one of the following statements is correct about the following reaction



- A. The reaction proceeds via free radical benzylic intermediate
 B. The reaction proceed via $\text{S}_{\text{N}}2$ reaction mechanism
 C. The reaction proceeds via $\text{S}_{\text{N}}1$ reaction mechanism (carbocation intermediate)
 D. The reaction could be good example aromatic nucleophilic substitution reaction by elimination-addition mechanism
 E. None of the above
32. What is the **major organic product (E)** of the sequences of reaction shown below?

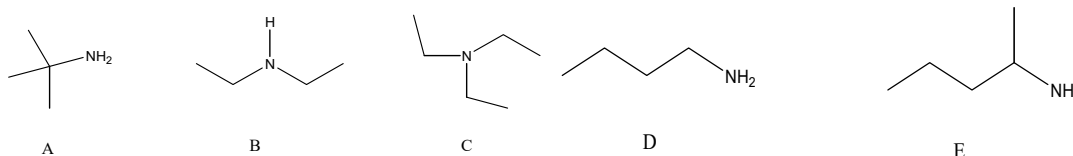


A.

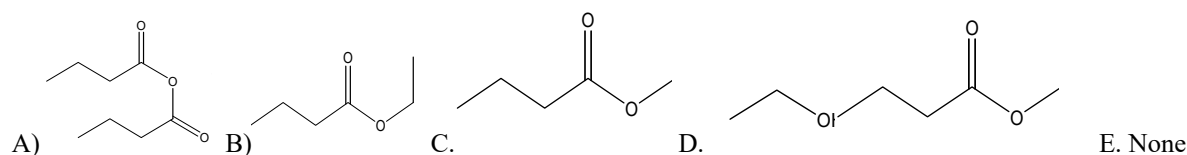
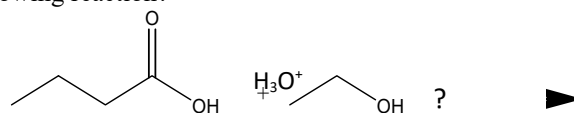
33. Identify the **correct** statement about **electrophilic aromatic substitution reaction**.

- A. All ortho-para directing groups always activate an aromatic ring and lead the incoming electrophile (E^+) to ortho- and/or para-positions of the ring
- B. ortho- and para directing substituent groups lead to the formation ortho- and para-substituted products as major products and meta-product as minor product
- C. Meta-directing groups do not favor attack of an electrophile at meta-position of the aromatic ring
- D. Some ortho-para directing groups can result in the formation of meta-substituted product(s) as major product(s) of electrophilic aromatic substitution reaction
- E. None of the above

34. Which of the following structure represents a tertiary amine?



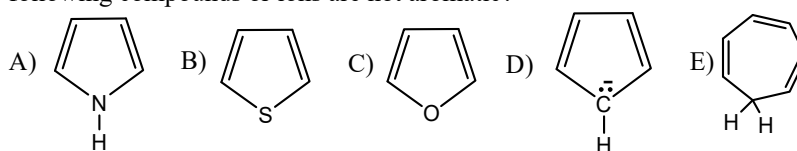
35. Identify product of the following reaction?



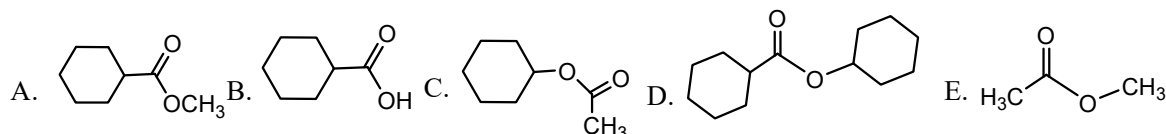
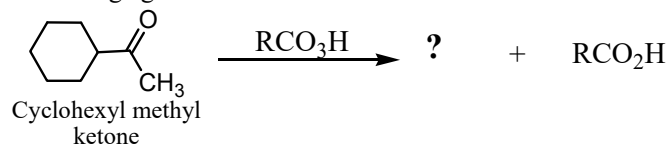
36. Which one of the following statement is **correct** about iodoform reaction (test)?

- A. All ketones can give positive result to this test
- B. This reaction transforms a methyl ketone into a carboxylic acid with one less carbon.
- C. 2-Butanone and 3-hexanone give positive test to iodoform reactions
- D. In such reactions, observation of a pale yellow solid (precipitate) is good indicator that the unknown ketone is not a methyl ketone
- E. None of the above

37. Which one of the following compounds or ions are not aromatic?



38. The **Baeyer-Villiger** oxidation is a process of oxidation of aldehydes and ketones into carboxylic acids and esters in the presence of peroxy acids (RCO_3H). What is the product of oxidation of cyclohexyl methyl ketone in the presence of peroxy acid as an oxidizing agent?



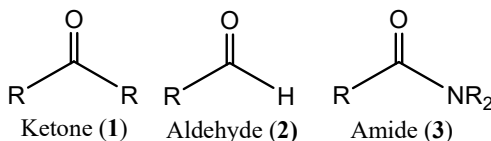
25. The following two amines (tetrahydroquinoline and tetrahydroisoquinoline) shown differ by a factor of 40,000 in their basicity strength (or K_b values). Which one of them a stronger base?



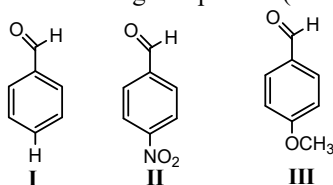
- A. tetrahydroquinoline
- B. Both compounds tetrahydroquinoline and tetrahydroisoquinolin have equal basic strength

- C. tetrahydroisoquinolin
- D. It is difficult to decide
- E. None of the above

39. Which one of the following statements is **true** about reduction of carbonyl compounds (**1-3**) by hydride reagents such as LiAlH_4 and NaBH_4 ?



- A. Both LiAlH_4 and NaBH_4 can reduce all the above carbonyl compounds by donating hydride ion (H^-) as nucleophile
 - B. LiAlH_4 can reduce all the compounds and the reactivity order is **2 > 1 > 3**
 - C. NaBH_4 can reduce only compound **3**
 - D. LiAlH_4 can reduce only compounds **1** and **2** but NaBH_4 reduce compounds **1-3**
 - E. None of the above
40. When the relative rates of reactivities of the following compounds (**I-III**) to the same nucleophile are compared;



- A. **I > II > III** B. **II > I > III** C. **II > III > I** D. **III > I > II** E. All have the same relative rate

41. Which one of the following statement is **correct** about nucleophilic addition reactions of carbonyl compounds such as ketones and aldehydes?

- A. Such reactions always need acid catalyst to add a nucleophilic species to electrophilic carbon atoms of carbonyl compounds
- B. When weak nucleophiles react with carbonyl compounds, addition initially takes place at the carbonyl carbon that is followed by protonation.
- C. When strong nucleophiles react with carbonyl compounds, protonation of the carbonyl oxygen takes place first and followed by attack of the nucleophile at the carbonyl carbon
- D. Both weak nucleophile and strong nucleophile follow the same mechanism (procedure) when they react ketones or aldehydes
- E. Propanal ($\text{C}_3\text{H}_6\text{O}$) is more reactive than 2-propanone ($\text{C}_3\text{H}_6\text{O}$) during nucleophilic addition reactions

30. Which one of the following methods is useful to prepare **esters from ketones** and **cyclic ketones to lactones**? A. Beckmann rearrangement B. Hofmann elimination C. Baeyer-Villiger oxidation D. Gabriel synthesis E. None of the above

31. Which one of the following statement is **wrong** about **Friedel-Craft's** alkylation aromatic compounds (e.g., **benzene**)?

- A. It gives polyalkylated products since an alkyl group once introduced into the ring, it makes the ring more susceptible or more reactive than benzene toward electrophilic aromatic substitution reaction
- B. Some compounds such as aniline do not undergo **Friedel-Craft's** alkylation
- C. Aromatic compounds with electron withdrawing substituent groups yield poor yield in **Friedel-Craft's** alkylation reactions
- D. In some cases, the alkyl group (**R-**) from the alkylating agent (**R-X**) may undergo rearrangement to give unexpected alkylated aromatic compounds as major products
- E. None of the above

32. Which of the following actinoids is used as a nuclear fuel?

- A. Actinium B. Thorium C. **Uranium** D. Californium

33. Which of the following statements is correct about lanthanoids and actinoids ?

- a. Both actinoids and lanthanoids are less basic
- b. Both actinoids and lanthanoids do not exhibit magnetic and spectral properties
- c. **Both actinoids and lanthanoids are electropositive**
- d. Both actinoids and lanthanoids do not exhibit same oxidation state of +3

34. $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ is green but $[\text{Ni}(\text{en})_3]^{2+}$ is blue. The color difference arises due to

- a. The difference in coordination number

- b. The difference in the absorption frequency in the visible region
 - c. The difference in geometry
 - d. The high magnetic behavior of $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$
35. In industrial processes, transition elements and their oxides are used as
- a. surfactants
 - b. insecticides
 - c. catalyst
 - d. herbicides
36. Which electron configuration could correspond to a d-block metal ion that is colorless in solution?
- a. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8$
 - b. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$
 - c. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^5$
 - d. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10}$
37. When KMnO_4 is added to oxalic acid solution, the decolorization is slow in the beginning but becomes instantaneous after sometimes because
- a. CO_2 is formed as a product
 - b. Reaction is exothermic
 - c. MnO_4^- catalyses the reaction
 - d. Mn^{2+} acts as autocatalyst
38. The value of 'spin only' magnetic moment for one of the following configurations is 2.84 BM. Which one is the correct one ?
- a. d^4 (in strong ligand field)
 - b. d^4 (in weak ligand field)
 - c. d^3 (in weak as well as strong fields)
 - d. d^5 (in strong ligand field)
39. Cesium ($Z = 58$) is an important member of the lanthanoids. Which of the following statement about cesium is incorrect?
- a. The common oxidation states of cesium are + 3 and + 4.
 - b. The + 3 oxidation state of cesium is more stable than + 4 oxidation state.
 - c. The + 4 oxidation state of cesium is not known in solutions.
 - d. Cesium (IV) acts as an oxidizing agent.
40. The electronic configuration of Cu(II) is $3d^9$ whereas that of electronic configuration of Cu(I) is $3d^{10}$. Which of the following is correct?
- a. Cu(II) is more stable
 - b. Cu(II) is less stable
 - c. Cu(I) and Cu(II) have equal stability
 - d. Stability of Cu(I) and Cu(II) depend on the nature of the salt
41. Gadolinium (Gd) belongs to 4f series. Its atomic number is 64. Which of the following is the correct electronic configuration of Gd^{2+} ?
- a. $[\text{Xe}] 4f^6 5d^1 6s^2$
 - b. $[\text{Xe}] 4f^7 5d^1$
 - c. $[\text{Xe}] 4f^8 6d^1$
 - d. $[\text{Xe}] 4f^9 5s^1$

42. Which one of the following is an inner orbital complex as well as diamagnetic in behaviour (Atomic number : Zn = 30), Cr = 24, Co = 27, Ni = 28)
- $[\text{Ni}(\text{NH}_3)_6]^{2+}$
 - $[\text{Cr}(\text{NH}_3)_6]^{3+}$
 - $[\text{Co}(\text{NH}_3)_6]^{3+}$
 - $[\text{Zn}(\text{NH}_3)_6]^{2+}$
43. Which one of the following characteristics of the transition metals is associated with higher catalytic activity?
- High enthalpy of atomisation
 - Paramagnetic behaviour
 - Colour of hydrate ions
 - Variable oxidation states
44. If $\Delta_o < P$, then the correct electronic configuration for d^5 system will be
- $t_{2g}^4 e_g^1$
 - $t_{2g}^2 e_g^3$
 - $t_{2g}^3 e_g^2$
 - $t_{2g}^5 e_g^0$
45. Which of the following complex salt / ion is incorrectly named according to the IUPAC rules ?
- $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ = Trioxalatoferrate (III) ion
 - $[\text{Co}(\text{NH}_3)_5 \text{ONO}]^{2+}$ = Pentamminenitrito –O-cobalt (III) ion
 - $[\text{Co}(\text{en})_3] \text{Cl}_3$ = Tris (ethylenediamine) Cobalt (III) Chloride
 - $[\text{Co}(\text{NH}_3)_5 \text{Cl}] \text{Cl}_2$ = Pentamminechlorido cobalt (II) di Chloride
46. The most abundant isotope of hydrogen is
- Tritium
 - Deuterium
 - Protium
 - Parahydrogen
47. Which of the following is wrong about hydrogen ?
- Hydrogen is the most abundant element in the universe and its chemistry is truly unique
 - Compounds of hydrogen are insoluble in water
 - It forms ionic hydrides with active metals
 - forms covalent compounds with an oxidation state of 1– with less electronegative elements
48. Identify among the following the incorrect statement about hydrogen? Hydrogen
- can be prepared by the reaction of less reactive metals with steam or by electrolysis of water
 - Has a strong force of attraction between its molecules.
 - acts as a reducing agent in its reaction with halogens, oxygen or sulfur
 - Is used in making nitrogenous fertilizers and synthesizing ammonia.
49. Which of the following atoms, designated by their electron configurations, has the highest first ionization energy?
- $[\text{Ne}] 3s^2 3p^1$
 - $[\text{Ne}] 3s^2 3p^3$
 - $[\text{Kr}] 4d^{10} 5s^2 5p^3$
 - $[\text{Xe}] 4f^{14} 5d^{10} 6s^2 6p^3$
50. Identify the correct statement
- Elemental sodium can be prepared and isolated by electrolyzing an aqueous solution of sodium chloride
 - Elemental sodium is a strong oxidizing agent
 - Elemental sodium is insoluble in ammonia
 - Elemental sodium is easily oxidized

51. Which of the following is NOT true for the Group 1A elements?
- A. Most of them are soft, silvery, corrosive metals.
 - B. Their atomic radii and ionic radii remain constant down the group
 - C. They are extremely reactive metals and usually produced by electrolysis
 - D. They are excellent conductors of heat and electricity.
52. The solubility of the alkali metal carbonates
- A. Increases at first and then decreases
 - B. Does not show regular variation
 - C. Increases as we go down the group
 - D. Decreases as we go down the group
53. Which of the following properties is not true for an alkali metal
- A. Low atomic volume
 - B. Low ionization energy
 - C. Low density
 - D. Low electronegativity
54. In the manufacture of metallic sodium by the fused salt electrolysis (Down's process) a small amount of calcium chloride is added to
- A. Improve the electrical conduction
 - B. Increase the temperature of electrolysis
 - C. Bring down the melt temperature
 - D. Stabilize the metallic sodium
55. Lithium shows similarities to magnesium in its chemical behaviour because
- A. Similar size, greater electronegativity and similar polarizing power.
 - B. Similar size, same electronegativity and lower polarizing power
 - C. Similar size, same electronegativity and similar high polarizing power
 - D. different size, less electronegativity but similar polarizing power
56. Which of the following order is correct with respect to the hydration enthalpy?
- A. $B > Mg > Ca < Sr > Ba$
 - B. $Be^{+2} > Mg^{+2} > Ca^{+2} > Sr^{+2} > Ba^{+2}$
 - C. $B > Mg < Ca > Sr > Ba$
 - D. $B > Mg > Ca > Sr < Ba$
57. Which of the following is a component of milk of magnesia?
- A. magnesium oxide
 - B. magnesium sulphate
 - C. magnesium hydroxide
 - D. magnesium chloride
58. Sample of barium and calcium are added separately to beakers of cold water which contain a few drops of litmus solution. Which observation will be made with only the calcium and not with the barium?
- A. A gas is evolved
 - B. A white suspension appears in water
 - C. The solution turns blue
 - D. Fast reaction takes place
59. Which statement about the general trend in melting points for the group IIA metals is correct?

A. The melting point increases as you move down group IIA because there is more delocalized electron in the metallic structure.

B. The melting point decreases as you move down group IIA because the force of attraction between the layers of positive and delocalized electrons is less

C. The melting point increases as you move down group IIA because more energy is required to break the metallic bonds

D. The melting point increases as you move down group IIA because the Vander Waals force is larger due to larger metal atoms within the structure

60. Which is the correct use of a group II A compound ?

A. MgO is commonly used in agriculture to reduce soil acidity

B. $\text{Ca}(\text{OH})_2$ is used as an in digestion remedy

C. SrCO_3 is used in fireworks and flares

61. D. BaCl_2 is used as barium meals in medicine

62. Sodium sulphate is soluble in water whereas barium sulphate is sparingly soluble because

A. The hydration energy of Na_2SO_4 is less than its lattice energy

B. The hydration energy of Na_2SO_4 is more than its lattice energy

C. The lattice energy of BaSO_4 is more than its hydration energy

D. The lattice energy has no role to play in solubility

63. Bond order of 1.5 is shown by

A. O_2^{2-}

B. O_2

C. O_2^+

D. O_2

64. In case of group III elements, the inert pair effect is predominant in

Note: Inert pair effect is the tendency of 's' electrons not to participate in bonding.

A. Boron

B. Aluminum

C. Indium

D. Thallium

65. Tetrachloride with the lowest boiling point is

A. SiCl_4

B. CCl_4

C. SnCl_4

D. PbCl_4

66. 22. In general a salt dissolves in water, when

A. hydration = lattice energy

B. hydration < lattice energy

C. hydration > lattice energy

D. There is no hydration energy

67. Bond dissociation enthalpy of E-H (E-elements) bonds is given below. Which of the compounds will act as strongest reducing agent?

Compound	NH_3	PH_3	AsH_3	SbH_3
$\Delta_{\text{diss}}(\text{EH})/\text{kJmol}^{-1}$	389	322	297	255

A. NH_3

B. PH_3

C. AsH_3

D. SbH_3

68. Which of the following statements wrong?

A. Single N-N bond is stronger than the single P-P bond

- B. PH_3 can act as a ligand in the formation of coordination compound with transition elements
- C. NO_2 is paramagnetic in nature
- D. Covalency of nitrogen in N_2O_5 is four
69. In case of nitrogen, NCl_3 is possible but not NCl_5 while in case of phosphorous, PCl_3 as well as PCl_5 are possible is due to
- A. availability of vacant d-orbital but not in N
- B. lower electronegativity of P than N
- C. lower tendency of H bond formation in P than N
- D. occur of P in solid while N in gaseous state at room temperature
70. Which of the following metallic bonding theories explain why some substances are electrical conductors, some are not, and some are semiconductors ?
- A. Molecular orbital or band theory
- B. Valence bond theory
- C. Electron sea model
- D. Crystal field theory
71. If the uncertainty in pH is equal to 5.21 ± 0.03 . What is the concentration of hydronium ion and its uncertainty?
- A. $6.2(\pm 0.2) \times 10^{-6} \text{M}$
- B. $6.2(\pm 0.4) \times 10^{-6} \text{M}$
- C. $6.2(\pm 0.6) \times 10^{-6} \text{M}$
- D. $6.2(\pm 0.7) \times 10^{-6} \text{M}$
72. What is the molarity of a solution of NaOH that has 0.491 g dissolved in 400.0 mL of solution?
- A. 0.03 M
- B. 0.20 M
- C. 0.30 M
- D. 0.02 M
73. Which of the following is false about the major requirements for reactions used in titration?
- A. Be fast and complete
- B. Have large equilibrium constant
- C. Occur in fixed, reproducible amounts
- D. It should be a second order reactions
74. Which of the following primary standard is used to determine the concentration of NaOH in a solution?
- A. Sodium carbonate
- B. Sodium tetraborate
- C. Potassium dichromate
- D. Potassium hydrogen phthalate
75. What is the volume of 0.2 N HCl solution required to react completely with 2.12 g of pure sodium carbonate, Na_2CO_3 ?
- A. 150 mL
- B. 250 mL
- C. 200 mL
- D. 400 mL
76. Which of the following statements concerning Mohr's method are correct? (1) It is a kind of titration used for silver-halide precipitation titrations (2) Potassium dichromate is used as the indicator (3) It should be carried out in pH between 6.5 and 10
- A. (1) and (2) only
- B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

77. How many liters of water must be added to 4 liters of a 6 M HNO_3 solution to create a solution that is 2 M?

A. 2 liters

B. 4 liters

C. 6 liters

D. 8 liters

78. What is the molality of a 10% (by weight) $\text{C}_6\text{H}_5\text{O}$ solution?

A. 0.012 m

B. 0.12 m

C. 1.2 m

D. 12 m

79. What is the volume of a 36.45% solution of hydrochloric acid (density = 1.50 g/mL) required to prepare 9.0 liters of a 5.0 M solution.

A. 1.0 liter

B. 2.0 liters

C. 2.5 liters

D. 3.0 liters

80. Which of the following best describes the pH of a 0.01 M solution of HBrO ? (For HBrO , $K_a = 2 \times 10^{-9}$)

A. Equal to 7

B. Between 2 and 7

C. Between 7 and 11

D. Greater than or equal to 11

81. What is the pH of a 0.1 M solution of ammonium chloride at 25 °C? (K_b for $\text{NH}_3 = 1.8 \times 10^{-5}$)

A. 11.13

B. 8.87

C. 5.13

D. 2.87

82. If 0.040 moles of solid NaOH is added to 1.0 liter of a solution that is 0.10 M in NH_3 ($K_b = 1.8 \times 10^{-5}$) and 0.20 M in NH_4Cl , what will be the pH of the resulting solution? Assume no volume change.

A. 4.80

B. 5.05

C. 8.65

D. 9.20

83. The amount of chloride in a water supply is determined by Mohr's titration. What percentage of chloride is present in the water if 26.7 mL of 0.30 M AgNO_3 is necessary to react with all of the chloride in a 14.0 grams sample?

A. 2.03%

B. 2.26%

C. 0.316%

D. 3.51%

84. If 200.0 mL of 1×10^{-7} M silver nitrate is mixed with 200.0 mL of 1×10^{-8} M sodium iodide, what will occur? For silver iodide, $K_{sp} = 8.3 \times 10^{-17}$.

A. Silver iodide will precipitate

B. Sodium nitrate will precipitate

C. Silver nitrate will precipitate

D. No precipitate will form

85. If you titrate 100 mL of a 0.1 M MgCl_2 solution by a 0.1 M AgNO_3 , then pCl at equivalence point equals ($K_{\text{sp}} \text{ AgCl} = 1.2 \times 10^{-10}$)
- A. 3.30
 - B. 4.96**
 - C. 7.60
 - D. 7.08
86. Calculate the molar solubility AgCl in 6.0 M NH_3 at a temperature of 298 K (K_{sp} for AgCl is 1.8×10^{-10} and K_f for $\text{Ag}(\text{NH}_3)_2^+$ is 1.7×10^7).
- A. 1.2998
 - B. 0.3319**
 - C. 1.1112
 - D. 0.0316
87. Calculate the concentration of free Zn^{2+} ion when 0.010 mol $\text{Zn}(\text{NO}_3)_2$ is added to 1.0 L of solution that is buffered at pH 12.0. K_f for $[\text{Zn}(\text{OH})_4]^{2-} = 2.9 \times 10^{15}$.
- A. 3.4×10^{-10} M**
 - B. 3.4×10^{-16} M
 - C. 3.4×10^{-18} M
 - D. 2.9×10^{-13} M

ANSWER: A

88. Calculate the pOH of a solution made by mixing 70 mL of 0.10 M NH_3 ($K_b = 1.8 \times 10^{-3}$) and 60 mL of 0.05 M HCl .
- A. 2.00
 - B. 2.62**
 - C. 2.87
 - D. 3.00

ANSWER: B

89. 20 mL of 0.1 M HF ($K_a = 6.8 \times 10^{-4}$) is titrated with 0.1 M NaOH . What is the pH of the solution when 10 mL of NaOH have been added?
- A. 2.1
 - B. 3.2**
 - C. 7.2
 - D. 6.3
90. Calculate the reduction potential at 25 °C for the conversion of $\text{MnO}_4^-(\text{aq})$ to $\text{MnO}_2(\text{s})$ in aqueous solution at pH = 9.0 and 1.0 M $\text{MnO}_4^-(\text{aq})$ given that $E^\circ (\text{MnO}_4^-/\text{MnO}_2) = +$
- A. 1.69 V.
 - B. 2.08 V**
 - C. 1.22 V
 - D. 2.43 V

Questions 27 and 28 are answered based on the following information: A common analytical experiment is the gravimetric analysis of copper in brass. Five samples were analyzed and the percentage of copper in each determined to be 93.42%, 93.86%, 92.78%, 93.14%, and 93.60% by mass.

91. What is the relative standard deviation of the result?
- A. 0.441
 - B. 0.447**
 - C. 0.271
 - D. 0.275
92. What is the 95% confidence interval for the percentage of copper in brass (use t value at 95% CL = 2.78 for 4 degrees of freedom)?
- A. $(93.4 \pm 0.5)\%$**

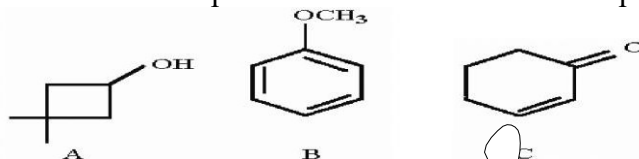
- B. $(93.9 \pm 0.5)\%$
 C. $(93.4 \pm 0.6)\%$
 D. $93.9 \pm 0.6)\%$
93. Four determinations of a bromide solution in a sample yielded the following values: 43.28%, 43.19%, 43.62%, and 43.24%. The Q-test value for four measurements at the 90% confidence level is 0.76. What is the best average to report for these results?
 A. 43.24%
 B. 43.33%
 C. 43.19%
 D. 43.28%
94. A sample of uranium-bearing material weighing 1.60 grams yielded 0.40 gram of UO_3 . What is the percentage of uranium (Atomic mass of U = 238.028 g/mol) in the sample?
 A. 25.0%
 B. 20.8%
 C. 29.5%
 D. 21.2%
95. Hydrocortisone sodium phosphate (prevent allergic reaction), A (1%, 1 cm) value=333 at 248 nm, measured absorbance=0.666 at 248 nm. What are the concentrations of the following solutions of drugs in g/100 ml and mg/100 ml?
 A. 1 and 0.001 B. 0.002 and 2 C. 1 and 0.666 D. 333 and 248
96. For the analysis of nutrients, the first step in doing experiments by using Atomic Absorption Spectroscopy is that
 a. Insert the HCL of the metal to be analysed B. Aspirate the blank solution
 b. Adjust the wavelength C. Aspirate the standards
 E. Prepare the sample (extraction, digestion, dilution)
97. The process of converting liquids in to fine droplets through pressure is:
 c. Atomization B. Nebulization
 C. Vaporisation D. Desolvation E. None
98. The sensitivity of Atomic Absorption and Atomic emission Spectroscopy rely on:
 d. the type of Atomizer
 e. the type of the Nebulizer
 f. the type of wave length selector
 g. the type of signal processer
 h. the nature of out put devices
99. According to Beer's law for a colored solution, which one of the following statements is False?
 A) The percent transmission is not directly proportional to the concentration.

- B) The absorbance is directly proportional to the concentration.
 C) The **absorbance** does not change when the cell path length (cell thickness) increases.
 D) When the concentration of the unknown sample is higher than the highest concentration in the calibration range, the sample should be diluted.

100. In UV absorption spectroscopy, which of the following statements are true?
 i. $\sigma \rightarrow \sigma^*$ transitions are lower in energy than $\pi \rightarrow \pi^*$ transitions
 j. A conjugated systems of double bonds in a molecular shifts the a absorption maxima to higher wavelengths
 k. $\sigma \rightarrow \sigma^*$ transitions are higher in energy than $\pi \rightarrow \pi^*$ transitions
 l. B **and** C
101. How should the concentration of a colorless sample be determined?
 m. **Using** a UV spectrophotometer
 n. Using a VIS spectrophotometer
 o. Either a UV or VIS spectrophotometer
 p. Using an indicator solution and pH paper
102. Which of the following relationships between absorbance and %Transmittance is incorrect ?
 q. $A = \log_{10} 100 / \%T$ b) $A = 2 - \log_{10} \%T$ c) $A = \log_{10} 1 / \%T$ d) None
103. A general term for science that deals with the interaction of various types of radiations with matter is ____?
 A. Photometry C. Chromatography.
 B. **Spectroscopy** D. polarimetry
 E. All of the above
104. A wave parameter which measure the number of oscillation of the field that occur per period is ____?
 A. Wave length C. **frequency** E. None of the above
 B. Amplitude D. wave number
105. Which of the following is an example of line source of radiation?
 A. D2 lamp B. Inert heated solids
 C. Hallow cathode lamp D. Laser E. **C&D**
106. From the given electromagnetic radiation the one having lowest wavelength and lowest Energy respectively are
 A. IR-radiation , Gamma-ray B. Radio wave, X-ray C. **Gamma** ray, radio wave
 D. IR- radiation, UV-radiation E. Visible radiation, Microwave
107. Calculate the wavelength and the energy associated with an NMR signal at 220MHZ. (Take $h=6.623 \times 10^{-34} Js$, $c = 3.00 \times 10^8 m/sec$)
 A. 2.727m, $2.13 \times 10^{-23} J$ B. 1.**36m**, $1.46 \times 10^{-25} J$ E. None
 C. 1.48m, $1.47 \times 10^{-24} J$ D. 2.65m, $1.23 \times 10^{-27} J$
108. From the given electromagnetic radiation the one having lowest wavelength and lowest Energy respectively are

- A. IR-radiation , Gamma-ray B. Radio wave, X-ray C. Gamma ray, radio wave
D. IR- radiation, UV-radiation E. Visible radiation, Microwave

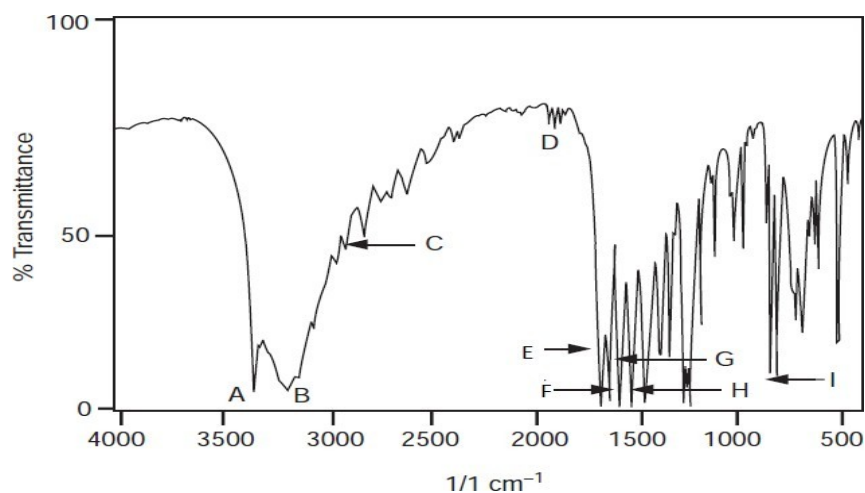
109. Among the following factor one **cannot** deviate from Beer's law
A. High concentration B. Stray light
C. monochromatic radiation D. solvent effect E. **none** of the above
110. The number of complete wavelength unit which pass a fixed point per unit of time is
A. Period B. **Frequency** C. Amplitude D. Intensity E. None
111. The purpose of **secondary** filter in fluorescence spectroscopy is
(A) Allows only excitation radiation
(B) **Allows** only emission radiation
(C) Allows both excitation and emission radiations
(D) Allows transmitted radiation
112. The selectivity and sensitivity of **fluorescence** spectroscopy is higher than that of absorption spectroscopy because of all of the following EXCEPT
(A) Fluorescence substance have different λ_{max} of excitation and emission
(B) Rigid molecules can only fluoresce
(C) Fluorescence intensity depends on the intensity of the incident radiation
(D) Every **molecule** that absorbs in UV region can show fluorescence
113. In fluorescence **spectroscopy**, emission spectra is obtained by keeping
(A) **Excitation** wavelength constant
(B) Emission wavelength constant
(C) Both excitation and emission wavelength constant
(D) Both excitation and emission wavelength varying
114. Which of the following is an example of line source of radiation?
A. D₂ lamp B. Inert heated solids
C. Hollow cathode lamp D. Laser E. **C&D**
115. The most widely used detector in spectrofluorometer is
(A) Barrier layer (B) Golay detector (C) Bolometer (D) **PMT**
116. Fluorescence is a result of transition of electron from
(A) Singlet ground state to singlet excited state
(B) **Lower** singlet excited state to singlet ground state.
(C) Triplet excited state to singlet ground state
(D) Triplet ground state to singlet excited state
117. The spectroscopy that measures the bond vibration frequencies in a molecule is
A) AAS C) NMR spectroscopy
B) **IR** spectroscopy D) Mass spectroscopy E) UV-Vis spectroscopy
118. Which compound would be expected to show intense IR absorption at 2820, 2710 and 1705 cm⁻¹?
A) CH₃COCH₂CH₃ B) CH₂=CHCOCH₃ C) PhCOCH₃ D) **PhCHO**
119. Which compound would be expected to show intense IR absorption at 1680 cm⁻¹?



D. all

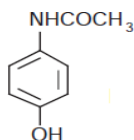
120. In a solid sample treatment technique, the finely ground solid sample is mixed with mineral oil(paraffin) to make a thick paste which is then spread between IR transmitting windows. What is the name of this solid sample treatment technique?
A) Pressed pellet B) **Mull** technique C) Solid films D) Solids run in solution
121. The following spectrum shows the output of IR instrument. Among the following drug

molecule, which of these molecules best corresponding to the IR spectrum below? The drugs are analyzed by IR as KBr discs.

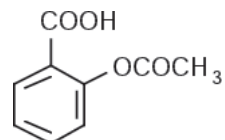


a. Infrared **spectrum** of *paracetamol* .

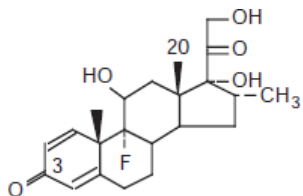
b. Infrared spectrum of *aspirin*.



c. Infrared spectrum of *dexamethasone*



d. All of the above



122. Which of the following source is continuous source for IR spectroscopy?

A) Deuterium discharge lamp

C) Tungsten lamp

B) **Nernst Glower**

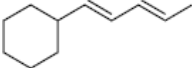
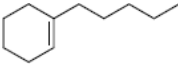
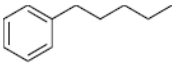
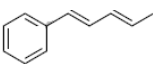
D) Hollow cathode lamp

123. A solution of caffeine in chloroform is provided for analysis, with the concentration of caffeine in the solution being required. Caffeine in chloroform shows a distinct carbonyl band at 1656 cm^{-1} in the infrared spectrum (recorded by using a 0.1 mm NaCl cell). The infrared information provided by standard solutions of caffeine in chloroform is listed below in Table . Given that the unknown sample produces an absorbance of 0.166 at 1656 cm^{-1} in its infrared spectrum, determine the concentration of this caffeine sample. Assume that there is no interference from other components in the sample. $Y = 0.0165x + 0.0134$

Table : Calibration data for caffeine solutions

Concentration (mg ml^{-1})	Absorbance at (1656 cm^{-1})
0	0.000
5	0.105
10	0.190
15	0.265
20	0.333

- A. 0.166 mg ml^{-1} B. 9 mg ml^{-1} C. 25 mg ml^{-1} D. none
124. An NMR transmitter consists of
 A. Frequency synthesizer, RF signal generator, transmitter controller, and receiver
 B. CPU, RF signal generator, transmitter controller, and RF amplifier
 C. Frequency synthesizer, RF signal generator, and transmitter controller
 D. **Frequency** synthesizer, RF signal generator, transmitter controller, and RF amplifier
125. If the number of protons or neutrons is even the spin of the nucleus will be which of the following?
 r. Integral spin b) Half integral spin c) **Zero** spin d) Positive spin
126. Which list below gives *only* spin active nuclei?
 a. ^1H , ^{13}C , ^{19}F b. ^2H , ^{12}C , ^{19}F c. ^1H , ^{12}C , ^{19}F d. ^1H , ^2H , ^{12}C
127. The ^{13}C NMR spectrum of a compound A contains two signals and in the ^1H NMR spectrum there is a singlet. Which compound is consistent with these data?
 a. Dichloromethane b. **Acetone**. c. Bromoethane d. Ethanol.
128. How many signals do you expect to see in the ^1H NMR spectra of 2-bromopropane ($(\text{CH}_3)_2\text{CHBr}$) and 1-bromopropane $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$?
 a. 1-Bromopropane: 2; 2-bromopropane: 2. c. 1-**Bromopropane**: 3; 2-bromopropane: 2.
 b. 1-Bromopropane: 3; 2-bromopropane: 3. d. 1-Bromopropane: 2; 2-bromopropane: 3.

129. How many equivalent "NMR" protons are in butanoic acid? A) 8 B) 2 C) 6 **D) 4**
130. What is the order of decreasing vibrational frequency for C-Cl, C-Br, C-C, C-O and C-H?
s. C-H, C-C, C-O, C-Cl, C-Br b) C-Cl, C-Br, C-C, C-H, C-O
 c) C-O, C-H, C-Br, C-Cl, C-C d) C-Br, C-Cl, C-C, C-O, C-H
131. In **spectroscopy**, the position shift of a peak or **signal** to longer **wavelength** is?
 a. Bathochromic shift b. hypsochromic shift c. blue shift d. all
132. An NMR transmitter consists of
 A. Frequency synthesizer, RF signal generator, transmitter controller, and receiver
 B. CPU, RF signal generator, transmitter controller, and RF amplifier
 C. Frequency synthesizer, RF signal generator, and transmitter controller
D. Frequency synthesizer, RF signal generator, transmitter controller, and RF amplifier
133. Which one of the following compounds is expected to have the longest wavelength absorbance (λ_{max})?
 A)  B)  C)  **D) **
134. In UV absorption spectroscopy, which of the following statements are true?
 B) $\sigma \rightarrow \sigma^*$ transitions are lower in energy than $\pi \rightarrow \pi^*$ transitions
 C) A conjugated systems of double bonds in a molecular shifts the a absorption maxima to higher wavelengths
 D) $\sigma \rightarrow \sigma^*$ transitions are higher in energy than $\pi \rightarrow \pi^*$ transitions
E) B and C E. all
135. How should the concentration of a colorless sample be determined?
 F) **Using** a UV spectrophotometer B) Using a VIS spectrophotometer
 C) Either a UV or VIS spectrophotometer D) Using an indicator solution and pH paper
 E) all
136. The dichromate ion absorbs light of wavelength close to 500 nm. Based on this information, what can you conclude?
 A) Solutions of the dichromate ion are colorless.
 B) The **dichromate** ion absorbs within the visible region.
 C) The dichromate ion absorbs in the ultraviolet region.

D) The dichromate ion absorbs outside the visible region.

137. How many vibrational modes do you expect for SO_2 ? A) 2 B) 3 C) 4 D) 9

138. After amino acids separated using planar chromatography, the color of the spot may be colorless. Based on this what you will do to visualize the spots?

A. Spraying Etylenediamine on the spot

B. Spraying Iodine on the spot

C. Spraying Sulfuric acid on the spot

D. Spraying ninhydrine on the spot

139. During potentiometric determination of ferrous ammonium sulfate using potassium dichromate why Sulfuric acid is added to the solution?

A. To hydrolyze ferrous sulphate into ferrous hydroxide

B. To neutralize the solution

C. Oxidation of ferrous ion to ferric ion is possible only in the presence of acid

D. To increase the amount of ferrous ion in a solution

140. Advantages of gradient separations over isocratic separation is

A. Ability to separate complex samples

B. Equilibrated and much simpler to use

C. They can have longer run times

D. None

141. During determination of caffeine and Theobromine by HPLC in reversed phase,

A. Theobromine elute first from the column

B. Caffeine elute first from the column

C. Caffeine elute after from the column

D. Both elute equally from the column

142. What would happen if we set the voltage higher than 100 V in determination of DNA using electrophoretic power supply?

A. The higher the voltage, delay the DNA movement through the gel

B. Cause distortion of DNA bands

C. Cause the DNA to move to negatively charged electrode

D. All

143. The main advantage of using saturated KCl solution in reference electrode is ____

A. The low cost of KCl from other salt

B. The ease of reaction with analyte during the titration process

C. The chloride concentration does not change as the liquid evaporates

D. The ease of cell construction

144. Paper chromatography is a separation technique that used to separate

A. Simple mixture

B. Complex mixture

C. Viscous mixture

D. Metals

145. Which one of the following is **false** regarding to potentiometric titrations?

A. The potential is measured under static condition

B. Liquid junction potential will influence the study

C. It is more sensitive in ionic strength

D. Need an instrument that has two electrodes

146. Which of the following make thin layer chromatography is more preferable over paper chromatography for separation?

A. The samples elute with the help of gravity

B. A choice between different mobile phases

C. A choice between different stationary phases

D. Separation is based on polarity difference

147. EMF rises steeply soon after the equivalent point during titration of ferrous ammonium sulfate using potassium dichromate. This is due to:

A. The reduction potential of $\text{Fe}^{3+}/\text{Fe}^{2+}$

B. The reduction potential of ferric ion and dichromate ion

C. The reduction potential of $\text{Cr}^{6+}/\text{Cr}^{3+}$

D. The effect of sulfuric acid

148. What must be done to the solid samples for it to be introduced into the column without using solid injection syringes in gas chromatography?

A. Introduced in hot-zone of the column

B. Dissolved in volatile liquids

C. Introduced using rotary sample valve

D. Introduced using sampling loops

149. If a mixture containing both X1 (isoelectric point at pH 6.1) and X2 (isoelectric point at pH 5.0) amino acid is buffered at pH 5.5 and then undergo electrophoresis separation, the result will be

A. X1 will move to positively charged electrode

B. X2 will move to positively charged electrode

C. X1 will move to negatively charged electrode

D. Both amino acids not going to move any wear

150. The potential of a hydrogen electrode at pH = 10 is

A. 0.591 V

B. 0.00 V

C. -0.591 V

D. -0.059 V

151. Normal Aluminum electrode coupled with normal hydrogen electrode gives an EMF of 1.66 volts. So the standard electrode potential of Aluminum is

A. 1.66 V

B. -0.83 V

C. -1.66 V

D. 0.83 V

152. Which one of the following is **false** regarding to requirements needed during the development of planner chromatography?

A. The paper should be lined 2 cm above the edge of the chromatography

B. The tank should be closed

C. The spot of the sample should be large

D. All

Answer: C

153. The beaker solution is gradually changed into green during the titration of ferrous ammonium sulfate using potassium dichromate. This is due to

A. The formation of $\text{Fe}(\text{SO}_4)_3$

B. The formation of K_2S_4

C. The formation of $\text{Cr}_2(\text{SO}_4)_3$

D. The formation of FeSO_4

154. Assume that we using silica stationary phase and toluene mobile phase in a chromatography system, and the elution times for benzene, tetrahydrofuran and ethanol were determined. Which analyte elute first?

A. Benzene

B. Toluene

C. Tetrahydrofurane

D. Ethanol

155. Which of the following things make potentiometric titration is more preferable than classical titration

A. Sensitivity to pH

B. sensitivity to ionic strength

C. Takes much longer time

D. used for colored solution

Answer: D

156. In the first step, two dimensional gel electrophoresis generates a series of bands by isoelectric focusing. In a second step, a strip of this gel is placed on another gel contains SDS, and electric current is again applied. in this step:

A. protein with similar isoelectric points become further separated according to their molecular weight

B. The individual bands becomes stained so that the isoelectric focusing pattern can be visualized

C. The individual bands become visualized by interacting with protein specific antibodies in the second gel

D. The individual bands undergo more intense isoelectric focusing

157. Which of the following is not a desirable feature of the ovens used in gas chromatography?

A. It must have a fast rate of heating

B. Power consumption should be kept low

C. It must have maximum thermal gradients

D. It should have proper insulation

158. Which cell will measure standard electrode potential of copper electrode?

A. $\text{Pt (s)} \mid \text{H}_2 \text{ (g, 0.1 bar)} \mid \text{H}^+ \text{ (aq., 1 M)} \parallel \text{Cu}^{2+} \text{ (aq., 1M)} \mid \text{Cu}$

B. $\text{Pt(s)} \mid \text{H}_2 (\text{g}, 1 \text{ bar}) \mid \text{H}^+ (\text{aq.}, 1 \text{ M}) \parallel \text{Cu}^{2+} (\text{aq.}, 2 \text{ M}) \mid \text{Cu}$

C. $\text{Pt(s)} \mid \text{H}_2 (\text{g}, 1 \text{ bar}) \mid \text{H}^+ (\text{aq.}, 1 \text{ M}) \parallel \text{Cu}^{2+} (\text{aq.}, 1 \text{ M}) \mid \text{Cu}$

D. $\text{Pt(s)} \mid \text{H}_2 (\text{g}, 1 \text{ bar}) \mid \text{H}^+ (\text{aq.}, 0.1 \text{ M}) \parallel \text{Cu}^{2+} (\text{aq.}, 1 \text{ M}) \mid \text{Cu}$

159. Electrophoresis cell consist of

A. Power pack and electrophoresis unit

B. Electrophoresis unit and DNA separator

C. Buffer chamber and electrophoresis unit

D. Gel, buffer chamber and power pack

160. Which of the following factors does not influence electrophoretic mobility?

A. Molecular weight

B. Shape of molecule

C. Size of molecule

D. Stereochemistry of molecule

161. What is the necessity for a pH-sensitive electrode to be appreciably covered by glass?

A. To make it hygroscopic

B. To keep it from breaking

C. For fulfillment of the structure of the electrode

D. None

162. The significance of first derivative and second derivative plot in potentiometric titration is –

A. To get additional information about the redox reaction

B. To get the voltage of reference electrode

C. To get the value of standard electrode potential

D. To get more accurate equivalence point in case of colored and dilute solution

163. Among the following applications for which the conductometry titration is not used?

A. To determine moisture

B. Purity of water

C. Ionic product of water

D. Precipitation titration

164. The limiting current in a linear sweep voltammogram is related to:

A. The reduction potential of the reference electrode

B. The point at which concentration polarization begins

C. The concentration of the analyte of interest

D. The reduction potential of water

165. The end point in the coulometric titrations is determined by

A. Spectroscopy

B. Biamperometry

C. Potentiometry

D. All of the above

166. In the coulometric titration of U^{4+} in the presence of excess Ce^{3+} , it was found to require 652 seconds to reach the equivalence point using a constant current of 100.0 mA. How many moles of U^{4+} were present?



A. 1.35×10^{-3} moles

B. 3.38×10^{-4} moles

C. 1.69×10^{-4} moles

D. 6.76×10^{-4} moles

167. In conductometric titration when KOH is titrated against mixture of H_2SO_4 and malonic acid, which one will be reacting first?
- A. Malonic acid
 - B. Sodium malonate
 - C. Disodium malonate
 - D. H_2SO_4**
168. During determination of dichromate by a coulometric redox titration, why is the level of solution in the counter electrode's cell maintained above the solution level in the working electrode's cell?
- A. To prevent the solution that contains the analyte from entering the counter electrode's cell
 - B. To prevent the oxidation of H_2O at the counter electrode
 - C. To prevent from the reaction of O_2 with Fe^{2+}
 - D. All**
169. The concentration of $\text{Cr}_2\text{O}_7^{2-}$ in a sample is determined by a coulometric redox titration using Fe^{3+} as a mediator and electro generated Fe^{2+} as the titrant. Why is it necessary to remove dissolved oxygen by bubbling N_2 through the solution?
- A. To protect the reduction of Fe^{3+}
 - B. To increase the amount of Fe^{3+}
 - C. For complete reaction between Fe^{2+} with $\text{Cr}_2\text{O}_7^{2-}$**
 - D. Increasing the time to reach the titration's endpoint
170. Which one of the following GC detector is Very selective for organic compounds with halogen substituents, nitro and some other oxygen-containing functional groups & non-destructive?
- A. TCD
 - B. ECD**
 - C. FID
 - D. FPD
171. Which one of the following GC detector is Universal, non-destructive; concentration detector, limited dynamic range.
- A. TCD**
 - B. ECD
 - C. FID
 - D. FPD
172. Which one of the following GC column is the most efficient?
- A. Packed
 - B. WCOT
 - C. FSOT**
 - D. SCOT

173. Which one of the following HPLC injection system has a fixed volume loop like 20 μ l or 50 μ l or more with two modes; Load position and Inject mode?
- A. Septum Injector
B. Stop flow Injector
C. Rheodyne injector
D. On column injector
174. Which of the following is the heart of high performance liquid chromatography?
- A. Column
B. detector
C. solvent reservoir
D. Injector
175. Which one of the following purification technique applied for separation of sparingly soluble solid organic compound at room temperature in a liquid solvent, but highly soluble at higher temperatures?
- A. Distillation
B. Liquid liquid extraction
C. Recrystallization
D. solid liquid extraction
176. The length of fractionating column corresponding to single evaporation and then condensation step in fractional distillation is.
- A Height of column
B. HETP
C. Plate count
D. Number of theoretical plates
177. In which one of the analytical technique the analyt needs to be extracted from a solid material sample rather than a liquid, because of it is not possible or necessary to dissolve the entire sample or because of interferences that may also be present?
- A. Soxhlet extraction
B. Recrystallization
C. liquid liquid extraction
D. D. Filtration
178. A method of purification of liquids contaminated with either dissolved solids or miscible liquids based on the fact that the contaminants have different boiling points and vapor pressures than the liquid to be purified is?
- A. Distillation
B. liquid liquid extraction
C. Recrystallization
D. solid liquid extraction
179. Which one of the following apparatus used in liquid liquid extraction separation mechanism?
- A. distiller
B. Soxhlet apparatus
C. Separatory funnel
D. Filter paper
180. A process used to separate solids from liquids or gases using a filter medium that allows the fluid to pass through but not the solid is:
- A. Distillation
B. Sedimentation

C. Recrystallization

D. Filtration

181. The process of allowing particles in suspension in water to settle down at the bottom under the effect of gravity is:

A. Distillation

C. Recrystallization

B. Sedimentation

D. Filtration

182. A type of chromatography in which the stationary phase is a liquid chemically bonded to the surface of a solid substrate (bonded phase chromatography), while the mobile phase is either a liquid or gas is

A. Size exclusion

C. ion exchange

B. Adsorption

D. Partitioning

183. Which one of the following solvent is not best choice as mobile phase for reversed phase chromatography?

A. petroleum ether

C. acetic acid

B. acetonitrile

D. ammonia

184. In which of the following type of chromatography, the stationary phase consists of finely divided solid particles packed inside a tube, but with no stationary liquid substance present to function as the stationary phase?

A. Size exclusion

C. ion exchange

B. Adsorption

D. Partitioning

185. After sample injection, which one of the sample component reaches first to the detector system of a chromatography instrument?

A. with High affinity to the stationary phase

C. with low affinity to the stationary phase

B. the one Moving slowly through the stationary phase

D. with low affinity to the mobile phase

186. Two chromatography peaks show retention times of 3.5 and 1.5 min ($t_m = 0.5$ min). The width at the base of the first peak is 0.25 min, and the width at the base of the second peak is 0.75 min. which one of the following is not true concerning the separation?

A. separation is completed

C. relative retention is 3

B. separation is poor

D. capacity factor for the first peak is 6

187. In which of the band broadening term that the analyt molecules do not all take the same path through a packed column, which means that they do not all reach the detector at the same time, hence peak broadening is observed.
- A. mass transfer term
B. bloating paper effect term
C. eddy diffusion term
D. longitudinal diffusion term
188. Which one of the following factor of band broadening caused by the effect of high thickness of the stationary phase, which causes establishing slow equilibrium of the analyte between the stationary and mobile phases?
- A. mass transfer term
B. bloating paper effect
C. eddy diffusion term
D. longitudinal diffusion term
189. Which one of the following is not the advantage of HPLC over GC?
- A. Analysis is at Room temperature
B. Analysis of thermally stable compounds
C. Solvents consuming is very high
D. sample is Non-destructed during operation of HPLC
190. Which one of the following is not the drawback of plate theory in terms of defining the chromatographic experimental variables, which in practise are seen to affect the chromatographic separation.
- A. Does not consider thickness of the stationary phase
B. Does not consider uniformity of the packing material
C. Assumes HETP is uniform throughout separation
D. Doesn't explain why separation is achieved and the shape of the peaks.
191. The retention time of a certain component on a particular 20 meter gas chromatography column is 3.2 min. The width at the base for the peak is 0.2 min. what is the height equivalent to a theoretical plate?
- A. 0.0020
B. 0.0049
C. 0.0032
D. 0.0075
192. Which substance is the reducing agent in the following reaction?
- $$4\text{H}^+(\text{aq}) + 2\text{Cl}^-(\text{aq}) + \text{MnO}_2(\text{s}) \rightarrow \text{Cl}_2(\text{g}) + \text{Mn}^{2+}(\text{aq}) + 2\text{H}_2\text{O}(\text{l})$$
- A. $\text{H}^+(\text{aq})$
B. $\text{Cl}^-(\text{aq})$
C. $\text{Mn}^{2+}(\text{aq})$
D. $\text{MnO}_2(\text{s})$
193. The unit of specific conductance is
- A. $\Omega^{-1}\text{cm}^{-1}$
B. $\Omega^{-1}\text{cm}$
C. $\Omega^{-1}\text{cm}^2$
D. $\Omega^{-1}\text{cm}^{-1}\text{mol}^{-1}$
194. Specific resistance of a solution expressed as
- A. $\frac{l}{RA}$
B. $\frac{Rl}{A}$

C. $\frac{RA}{l}$

D. $\frac{Gl}{A}$

195. For the following electrochemical cell reaction shown, what is the standard free energy change ΔG° if $E^\circ = +1.61$ V?; $\text{Mg} / \text{Mg}^{2+}(\text{aq}) // \text{Zn}^{2+}(\text{aq}) / \text{Zn}$, $E^\circ = +1.61$ V

A. -311 kJ/mol

C. -155 kJ/mol

B. +311 kJ/mol

D. +155 kJ/mol

196. Given the electrochemical reaction shown, if the standard oxidation potential of Ni/Ni^{+2} is +0.26 V, and the standard reduction potential of Al^{3+}/Al is -1.66V, what is E° value?

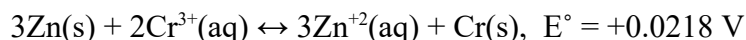
A. +1.26 V

C. +0.46 V

B. +0.85 V

D. +1.40 V

197. Electrochemical cell potentials can be used to determine equilibrium constants that would be otherwise difficult to determine because concentrations are small. What is K for the following balanced reaction, if $E^\circ = +0.0218$ V?



A. 1.3×10^{-3}

C. 162

B. 2.2

D. 0.37

198. The conductance of a 1cm^3 volume solution, when the two electrodes of a conductometer are separated by unit distance, 1cm is termed as

A. Specific resistance

C. Equivalent conductance

B. Molar conductance

D. Specific conductance

199. The value of E° for the following reaction is 1.260 V. What is the value of E_{cell} ? concentrations are given: $[\text{Cd}^{2+}] = 0.1\text{M}$ & $[\text{Al}^{3+}] = 0.6\text{M}$



A. 1.235 V

C. 1.15 V

B. 1.285 V

D. 1.37 V

200. The current we seek to measure in voltammetry due to reduction or oxidation of analyte at the working electrode.

A. Faradaic current

C. Charging current

B. Limiting current

D. Residual current

201. The peak current measured in reversible cyclic voltammetric analysis depends on

A. Type of analyte

C. Concentration & scan rate

B. pH of the solution

D. All

202. Which one of the following mass transport processes occurs when charged particles in solution are attracted or repelled from an electrode that has a positive or negative surface charge?

A. Diffusion

C. Migration

B. Convection

D. Stirring

203. In polarographic wave form the potential at which the working electrode is capable of reducing & analyte current starts to be registered at the electrode is known as

A. Half wave potential

C. decomposition potential

B. Applied potential

D. Standard reduction potential

204. The resistance of 0.2N KCl solution is found 500 ohm, when measured in a conductivity cell. If the specific conductance of 0.2 N KCl is 0.12504, calculate the equivalent conductance.

A. $200 \text{ ohm}^{-1}\text{cm}^2/\text{equivalent}$

C. $125.0 \text{ ohm}^{-1}\text{cm}^2/\text{equivalent}$

B. $0.02 \text{ ohm}^{-1}\text{cm}^2/\text{equivalent}$

D. $625.2 \text{ ohm}^{-1}\text{cm}^2/\text{equivalent}$

205. The conductance of acetic acid at 25 °C is 16.3 ohm⁻¹ cm²/eq. The ionic conductance's of H⁺ and CH₃COO⁻ are 349.83 and 40.89 ohm⁻¹cm²/eq respectively. What is α for CH₃ COOH?

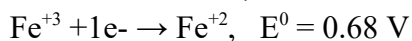
A. 0.04

C. 0.39

B. 0.03

D. 0.02

206. In the titration of 50 ml of 0.05 M Fe⁺² with 0.1 M Ce⁺⁴ in medium that is 1.0 M in H₂SO₄ at all times (assume the reference electrode is SHE) and the following standard potentials are given:



then what is the value of the potential of the system after adding 5 ml of Ce⁺⁴?

A. 0.64

B. 0.68V

C. 1.44V

D 2.12V

207. In question "above" what will be the value of the potential of the system after adding 25.1 ml of Ce⁺⁴?

A. 1.44v

B. 1.06v

C. 1.30v

D. 0.76

208. Λ^∞ m for NaCl, HCl and CH₃COONa are 126.4, 425.9 and 91.05 cm²mol⁻¹ respectively. Calculate Λ^∞ m for CH₃COOH.

A. 461.1 S cm²/mol

C. 389.5 S cm²/mol

B. 643.4 S cm²/mol

D. 350.5 S cm²/mol

209. The limiting current in a linear sweep voltammogram is related to:

A. The standard reduction potential for the redox couple under investigation

B. The reduction potential of the reference electrode

C. The concentration point at which of concentration the analyte of polarization interest begins

D. The reduction potential of water

210. What is the potential value in mV of the separation between the anodic and cathodic peaks of a reversible one electron cyclic voltammogram?

A. E_{1/2}

C. 59mv

B. (E_p + E_c)/2

D. Id/2

211. UV-visible spectrophotometer uses a combination of two lamps; which one is employed in the visible region?

A quartz

C deuterium

B tungsten

D plastic

212. Snell's law is related to

A. Refractometer

C. spectroscopy

B. Potentiometry

D. polarimeter

The complex Fe(SCN)⁺² has a molar absorptivity of 7x10³ Lmol⁻¹cm⁻¹ at 580 nm. Answer question 143-145 based on this information.

213. What is the absorbance of 2.5x10⁻⁵ M of the complex in 1cm cell?

A. 0.282

C. 0.095

B. 0.175

D. 0.357

214. What is the transmittance of the solution in question number 4?

A. 0.67

C. 0.54

B. 0.45

D. 0.76

215. The measured angle of rotation in polarimetry measurement depends upon

A. nature of sample

C. The length of the sample tube

B. Concentration

D. All of them

216. Which one of the following is the application of refractometry?

A. To check the purity of our sample

C. To determine unknown concentration

B. To know the identity of our sample

D. All of them

217. Which one of the following is the basic principle of polarimetry?

A. absorption

C. refraction of light

B. scattering of light

D. rotation of plane polarized light

218. How should the concentration of a red sample, such as hemoglobin, be determined?

A. Using a UV spectrophotometer

B. Using a VIS spectrophotometer

C. Either a UV or VIS spectrophotometer

D. With pH meter

219. Which one of the following is the importance of polarimetry?

A. Qualitative application

C. identification of sugar type

B. Quantitative application

D. all

220. Which one of the following is the basic principle of refractometry?

A. rotation of plane polarized light

C. refraction of light

B. scattering of light

D. absorption

221. Which one of the following spectroscopic technique does not need source of radiation?

A. Flame photometry

B. Atomic absorption spectroscopy

C. Uv vis spectroscopy

D. IR spectroscopy

222. Which one of the following electronic transition would be decreased λ_{max} value in polar solvent during uv vis absorption of a molecule?

A. $\pi \rightarrow \pi^*$

C. $\sigma \rightarrow \sigma^*$

B. $n \rightarrow \pi^*$

D. $\pi \rightarrow \sigma^*$

223. The sample cell of the flame atomic absorption spectroscopy system is

A. Detector

C. hollow cathode lamp

B. wave length selector

D. atomizer

224. The process of spray or fine droplet formation during the atomization process in FAAS is
- A. Ionization
B. Disolvation
C. Volatilization
D. Nebulization

225. Which electronic transitions Needs low energy during interaction of uv vis radiation?
- A. $\pi \rightarrow \pi^*$
B. $\sigma \rightarrow \sigma^*$
C. $n \rightarrow \sigma^*$
D. $n \rightarrow \pi^*$

226. A wave length selector optical component of uv visible instrument is known as
- A. Sample holder
B. Detector
C. Monochromator
D. light source

227. In uv vis absorption of a chromophor, If the solvent is polar, $\pi - \pi^*$ electronic transition shifts are?
- A. Hypso chromich shift
B. Red shift
C. Blue shift
D. A and C

228. The observed optical rotation value of a certain sugar solution by polarimeter with a sample Cell of 1dm is $+11^\circ$ and its specific rotation $[\alpha]_{590nm}^{20}$ is $+52.7^\circ$, Calculate the concentration of the Solution (g/ml).
- A. 0.21
B. 0.25
C. 4.8
D. 3.8

229. which type of Flame produce optimum temperature for atomization process in flame atomic absorption Spectroscopy?
- A. H_2/O_2
B. H_2/Air
C. C_2H_2/Air
D. C_2H_2/O_2

230. Beer Lambert's law gives the relation between which of the following?
- A. Reflected radiation and concentration
B. Scattered radiation and concentration
C. Energy absorption and concentration
D. Energy absorption and reflected radiation

231. The refractive index of CCl_4 at $20^\circ C$ is 1.46, if its density at this temperature is 1.60 g/cm^3 , what is the molar refraction of CCl_4 ?
- A. 0.37
B. 0.24
C. 26.36
D. 12.34

232. The representation of Beer Lambert's law is given as $A = abc$. If 'b' represents distance, 'c' represents concentration and 'A' represents absorption, what does 'a' represent?
- A. Intensity
B. Transmittance
C. Absorptivity
D. Admittance

233. Which electronic transitions are shown by only saturated hydrocarbons in far uv regions
- A. $\pi \rightarrow \pi^*$
B. $\sigma \rightarrow \sigma^*$
C. $n \rightarrow \sigma^*$
D. $n \rightarrow \pi^*$

234. A chemical reaction has reached equilibrium when
- A. the concentrations of reactants and products are equal.
 - B. all reactants have been converted to products.
 - C. all products have been removed from the reaction mixture.
 - D. the catalyst has been used up.
 - E. **the rate of the forward reaction equals the rate of the reverse reaction.**

235. The reaction for the decomposition of PCl_5 to chlorine and PCl_3 is shown below.



If the equilibrium concentrations are $[\text{PCl}_5] = 1.0 \text{ M}$, $[\text{PCl}_3] = 0.10 \text{ M}$, $[\text{Cl}_2] = 0.10 \text{ M}$, what is the value of the equilibrium constant?

- A. **1.0×10^{-2}**
- B. 1.0×10^{-4}
- C. 10×10^{-2}
- D. 1.0×10^2
- E. 2.0×10^{-2}

236. The equilibrium constant for the reaction for the decomposition of PCl_5 to chlorine and PCl_3 is 0.042.



If the equilibrium concentrations are $[\text{PCl}_3] = 0.010 \text{ M}$, $[\text{Cl}_2] = 0.10 \text{ M}$, what is the value of $[\text{PCl}_5]$?

- A. 0.010 M
- B. 0.0020 M
- C. 0.042 M
- D. **0.024 M**
- E. 0.0010 M

237. The equilibrium constant for the production of carbon dioxide from carbon monoxide and oxygen is

$K_c = 2 \times 10^{11}$. This means that the reaction mixture at equilibrium is likely to consist of

- A. mostly starting materials.
- B. an equal mixture of products and reactants.
- C. twice as much starting material as product.
- D. twice as much product as starting material.
- E. **mostly products.**

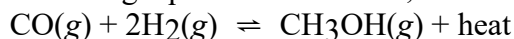
238. The value of the equilibrium constant for the combination of nitrogen and oxygen to make NO is 2×10^{-9} . What does this tell you about the concentrations of materials in the equilibrium mixture?

- A. The concentration of products exceeds the concentration of reactants.
- B. The concentrations of reactants and products are equal.
- C. The reactants are solids.
- D. **The concentration of reactants exceeds the concentration of products.**
- E. The products are solids.

239. Carbon monoxide binds to hemoglobin 140 times more strongly than oxygen does. What does this tell you about the equilibrium constants for the two reactions of hemoglobin with carbon monoxide and oxygen?

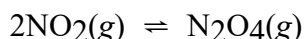
- A. **The equilibrium constant for the binding of CO is greater.**
- B. The equilibrium constant for the binding of oxygen is greater.
- C. The concentration of carbon monoxide at equilibrium is twice that of oxygen.
- D. Oxygen and carbon monoxide have the same formula mass.
- E. Oxygen and carbon monoxide react with hemoglobin in different fashions.

240. For the following equilibrium reaction, which cause and effect are correctly matched?



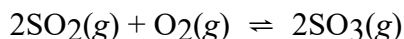
- A. add heat, shift right
- B. add CO, shift left
- C. remove CH₃OH, shift left
- D. remove heat, no change
- E. **remove H₂, shift left**

241. In the following gas phase reaction, what is the effect of adding more NO₂ to the starting reaction mixture?

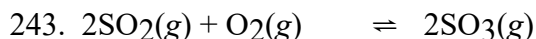


- A. It would make the reaction more endothermic.
- B. It would make the reaction more exothermic.
- C. It would slow the reaction down.
- D. It would decrease the final quantity of products.
- E. **It would increase the final quantity of products.**

242. In the following gas phase reaction, what is the effect on the direction of the reaction if more SO₃ is added to the reaction mixture?



- A. The equilibrium shifts to produce more products.
- B. The position of the equilibrium remains unchanged.
- C. The rate of formation of products is increased.
- D. **The equilibrium shifts to produce more reactants.**
- E. The catalyst for the reaction is used up.



For the above reaction at equilibrium, if O₂ is added, the amount of SO₂ present will

- A. **decrease.**
- B. increase.
- C. stay the same.
- D. None of the above

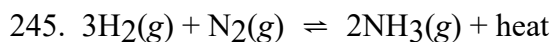
244. For the following reaction, the equilibrium constant K_c is 2.0 at a certain temperature.

Bromine can be liquefied easily and removed from the reaction vessel as it is formed. If this is done, how will it affect the equilibrium reaction?



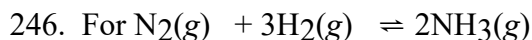
- A. **More products will be made as Br₂ is removed.**
- B. There will be a larger proportion NOBr in the vessel when equilibrium is reached.

- C. Less NO will be made.
- D. The pressure in the vessel will increase.
- E. The equilibrium constant will change.



For the above reaction at equilibrium, if the temperature is raised, the amount of N_2 will

- A. decrease.
- B. **increase.**
- C. stay the same.
- D. None of the above



- A. $K_c = K_p$
- B. $K_p = K_c (RT)$
- C. **$K_p = K_c (RT)^{-2}$**
- D. $K_p = K_c (RT)^{-1}$

247. Which of the following processes should have $\Delta S < 0$?

- A. $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$
- B. $\text{H}_2\text{O}(\text{s}) \rightarrow \text{H}_2\text{O}(\text{g})$
- C. **$2\text{NO}_2(\text{g}) \rightarrow \text{N}_2\text{O}_4(\text{g})$**
- D. $\text{NaCl}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{NaCl}(\text{aq})$

248. The equation for the standard enthalpy of formation of N_2O_3 is

- A. $\text{N}_2\text{O}(\text{g}) + \text{O}_2(\text{g}) \rightarrow \text{N}_2\text{O}_3(\text{g})$
- B. $\text{N}_2\text{O}_5(\text{g}) \rightarrow \text{N}_2\text{O}_3(\text{g}) + \text{O}_2(\text{g})$
- C. $\text{NO}(\text{g}) + \text{NO}_2(\text{g}) \rightarrow \text{N}_2\text{O}_3(\text{g})$
- D. **$\text{N}_2(\text{g}) + \frac{3}{2} \text{O}_2(\text{g}) \rightarrow \text{N}_2\text{O}_3(\text{g})$**

249. A mixture of two moles of carbon monoxides and one mole of oxygen, in a closed vessel is ignited to convert the carbon monoxide to carbon dioxide. If ΔH is the enthalpy change and ΔE is the change in internal energy, then

- A. $\Delta H < \Delta E$
- B. $\Delta H > \Delta E$
- C. **$\Delta H = \Delta E$**
- D. the relationship depends on the capacity of the vessel

250. When two solutions react the container "feels hot." Thus,

- A. the reaction is endothermic.
- B. **the reaction is exothermic.**
- C. the energy of the universe is increased.
- D. the energy of both the system and the surroundings is decreased.

251. Which statement about the effect of temperature on a reaction is incorrect?

- A. The rate constant is not directly proportional to the temperature.
- B. The rate constant depends on temperature.
- C. The rate of reaction depends on temperature.
- D. **The value of E_a is temperature dependent.**

252. Which one of the following changes would alter the rate constant (k) for the reaction $2A + B \rightarrow \text{products}$?

- A. increasing the concentration of A
- B. increasing the concentration of B
- C. increasing the temperature
- D. measuring k again after the reaction has run for a while

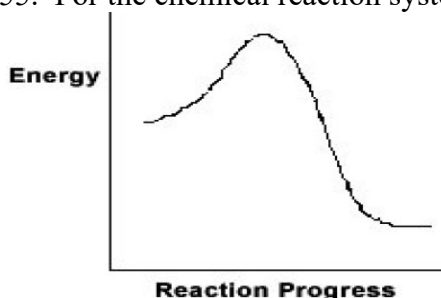
253. The half-life of a first-order reaction _____.

- A. is constant
- B. is the time necessary for the reactant concentration to drop to half its original value
- C. can be calculated from the reaction rate constant
- D. does not depend on the initial reactant concentration
- E. All of the above are correct.

254. For what order reaction does the half-life get longer as the initial concentration increases?

- A. zeroth order
- B. first order
- C. second order
- D. none of them because half-life is always independent of the initial concentration

255. For the chemical reaction system described by the diagram below, which statement is true?



- A. The forward reaction is endothermic.
- B. The activation energy for the forward reaction is greater than the activation energy for the reverse reaction.
- C. At equilibrium, the activation energy for the forward reaction is equal to the activation energy for the reverse reaction.
- D. The activation energy for the reverse reaction is greater than the activation energy for the forward reaction.
- E. The reverse reaction is exothermic.

256. An increase in the temperature of the reactants causes an increase in the rate of reaction.

The best explanation for this behavior is that as the temperature increases,

- A. the concentration of reactants increases.
- B. the activation energy decreases.
- C. the collision frequency increases.
- D. the fraction of collisions with total kinetic energy $> E_a$ increases.
- E. the activation energy increases.

257. According to the collision theory, all collisions do not lead to reaction. Which choice gives both reasons why not all collisions between reactant molecules lead to reaction?

1. The total energy of two colliding molecules is less than some minimum amount of energy.
2. Molecules cannot react with each other unless a catalyst is present.
3. Molecules that are improperly oriented during collision will not react.
4. Solids cannot react with gases.

- A. 1 and 2
- B. 1 and 3
- C. 1 and 4
- D. 2 and 3
- E. 3 and 4

258. Which one of the following statements concerning rates of reactions is FALSE?

- A. The higher the activation energy barrier, the faster the reaction.
- B. Increasing the concentration of a reactant may increase the rate of a reaction.
- C. Adding a catalyst speeds up the rate of reaction for both the forward and reverse reactions.
- D. Increasing the concentration increases the rate of a reaction, because it increases the number of collisions.

259. From the following statements which one is false about oxidizing agent?

- A. It is the species that gains electrons
- B. It is the species that is reduced
- C. It causes oxidation
- D. It is the species that loses electrons.

260. Consider the following reaction: $3\text{I}_2 + 3\text{H}_2\text{O} \rightarrow 6\text{H}^+ + 5\text{I}^- + \text{IO}_3^-$

In this reaction, the I_2 atoms undergo:

- A. oxidation only
- B. both oxidation and reduction
- C. reduction only
- D. neither oxidation nor reduction

261. An electrolytic cell uses electrical energy to drive

- A. chemical reaction
- B. physical reaction
- C. no reaction
- D. none of above

262. The purpose of the salt bridge in an electrochemical cell is to _____.

- A. provide a source of ions to react at the anode and cathode.
- B. provide oxygen to facilitate oxidation at the anode.
- C. provide a means for electrons to travel from the anode to the cathode.
- D. provide a means for electrons to travel from the cathode to the anode.
- E. maintain electrical neutrality in the half-cells via migration of ions.

263. In a voltaic cell, electrons flow from the _____ to the _____.

- A. salt bridge, anode
- B. anode, cathode
- C. anode, salt bridge
- D. salt bridge, cathode
- E. cathode, anode

264. The relationship between the change in Gibbs free energy (ΔG) and the emf (E) of an electrochemical cell is given by _____. n = mole, F = Faraday constant, T = temperature, R = universal gas constant

- A. $\Delta G = -E/nF$

- B. $\Delta G = -nRTF$
- C. $\Delta G = -nE$
- D. $\Delta G = -nF/ERT$
- E. $\Delta G = -nFE$

265. What happens to the reducing agent in an oxidation-reduction reaction?

- A. It is oxidized as it gains electrons.
- B. It is reduced as it gains electrons.
- C. It is oxidized as it loses electrons.
- D. It is reduced as it loses electrons.

266. The cell potential, E° , for an oxidation-reduction reaction was found to equal 1.10 V. What can be said about this reaction?

- A. at equilibrium
- B. endothermic
- C. nonspontaneous
- D. spontaneous

267. In Galvanic cell, electrons travel in which direction?

- A. from the anode to the cathode through the external circuit
- B. from the anode to the cathode through the porous cup
- C. from the cathode to the anode through the external circuit
- D. from the cathode to the anode through the porous cup.

268. From the following statements which one is false about half-reaction?

- A. Half reaction shows transferred electrons in the chemical equation.
- B. In oxidation half-reaction electrons are shown on the product side of the equation.
- C. In reduction half-reaction electrons are shown on the reactant side of the equation
- D. None of the above

269. Which of the following descriptions best describes an induced fit model?

- A. The process by which an active site alters shape such that it is ready to accept a substrate.
- B. The process by which a substrate adopts the correct binding conformation before entering an active site.
- C. The process by which a substrate binds to an active site and alters the shape of the active site.
- D. The process by which an active site alters the shape of the substrate such that it can adopt the necessary active conformation for binding.

270. Which of the following statements about enzymes is NOT true?

- A. Enzymes work best at a specified pH.
- B. All enzymes work inside cells.
- C. Enzymes are proteins.
- D. Enzymes are organic catalysts.

Answer: B

271. In IR spectrum the wave number range $1600\text{--}3500\text{ cm}^{-1}$ indicates?

A. fingerprint region

B. region used for functional group determination

C. the band due to complex stretching

D. All